A guided mathematics example

1 Define the notation

Let $q \in \mathbb{R}$ denote a nonnegative output quantity. The unit price is $p \geq 0$. The cost parameters satisfy $a > 0$ and $b > 0$. This is a hypothetical model for practising notation.

2 Group the related equations

Revenue and cost are given by

$$\operatorname{Rev}(q) = pq, \tag{1a}$$
$$\operatorname{Cost}(q) = aq + bq^2. \tag{1b}$$

System (1) contains the revenue equation (1a) and the cost equation (1b).

Define profit $\Pi(q)$ as revenue minus cost. One derivation has one number:

$$\begin{aligned} \Pi(q) &= \operatorname{Rev}(q) - \operatorname{Cost}(q) \\ &= pq - (aq + bq^2) \\ &= (p - a)q - bq^2. \end{aligned} \tag{2}$$

The equals signs in (2) line up at the alignment markers.

3 Write a boundary condition

For an interior optimum the derivative satisfies

$$\Pi'(q) = p - a - 2bq = 0.$$

Since $\Pi''(q) = -2b < 0$ and $q \geq 0$, the maximizing quantity is

$$q^* = \begin{cases} \dfrac{p - a}{2b} & \text{if } p > a, \\ 0 & \text{if } p \leq a. \end{cases}$$

The two conditions cover the boundary $p = a$ exactly once.

Proposition 3.1. *For every $q \geq 0$, the cost in* (1b) *is nonnegative.*

Proof. The assumptions imply $aq \geq 0$ and $bq^2 \geq 0$. Their sum is nonnegative. □

Proposition 3.1 follows from the stated signs, rather than from any typesetting command.